

Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie, Sharan Narang, Sharath Rapparthi, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Conguet, Virginie Do, Vish Vogeti, Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuwei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie DelPierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippos Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabsa, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangrabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny

- Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models. *arXiv*, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Alexandre Sallinen, Antoni-Joan Solergibert, Michael Zhang, Guillaume Boyé, Maud Dupont-Roc, Xavier Theimer-Lienhard, Etienne Boisson, Bastien Bernath, Hichem Hadhri, Antoine Tran, Tahseen Rabbani, Trevor Brokowski, Meditron Medical Doctor Working Group, Tim G. J. Rudner, and Mary-Anne Hartley. Llama-3-meditron: An open-weight suite of medical LLMs based on llama-3.1. In *Workshop on Large Language Models and Generative AI for Health at AAAI 2025*, 2025. URL <https://openreview.net/forum?id=ZcD35zKuj0>.
- Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin Dong, Yejin Choi, Jan Kautz, and Yi Dong. Prorl: Prolonged reinforcement learning expands reasoning boundaries in large language models. *arXiv preprint*, 2025. URL <https://arxiv.org/abs/2505.24864>.
- Ning Ding, Yulin Chen, Bokai Xu, Yujia Qin, Zhi Zheng, Shengding Hu, Zhiyuan Liu, Maosong Sun, and Bowen Zhou. Enhancing chat language models by scaling high-quality instructional conversations. *arXiv preprint arXiv:2305.14233*, 2023.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Tianle Li, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zhuohan Li, Zi Lin, Eric P. Xing, Joseph E. Gonzalez, Ion Stoica, and Hao Zhang. Lmsys-chat-1m: A large-scale real-world llm conversation dataset. *arXiv*, 2024. URL <https://arxiv.org/abs/2309.11998>.
- Xiangyu Qi, Ashwinee Panda, Kaifeng Lyu, Xiao Ma, Subhrajit Roy, Ahmad Beirami, Prateek Mittal, and Peter Henderson. Safety alignment should be made more than just a few tokens deep. *arXiv*, 2024. URL <https://arxiv.org/abs/2406.05946>.
- Joshua Engels, Logan Riggs, and Max Tegmark. Decomposing the dark matter of sparse autoencoders. *arXiv*, 2024. URL <https://arxiv.org/abs/2410.14670>.
- Chak Tou Leong, Qingyu Yin, Jian Wang, and Wenjie Li. Why safeguarded ships run aground? aligned large language models’ safety mechanisms tend to be anchored in the template region. *arXiv*, 2025. URL <https://arxiv.org/abs/2502.13946>.
- Leo Gao, Tom Dupre la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=tcsZt9ZNKD>.
- Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermy, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.
- Aleksandar Makelov, Georg Lange, and Neel Nanda. Towards principled evaluations of sparse autoencoders for interpretability and control. In *ICLR 2024 Workshop on Secure and Trustworthy Large Language Models*, 2024. URL <https://openreview.net/forum?id=MHIX9H8aYF>.
- Jacob Dunefsky, Philippe Chlenski, and Neel Nanda. Transcoders find interpretable LLM feature circuits. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. URL <https://openreview.net/forum?id=J6zHcScAo0>.

- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.
- Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. In D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 29, 2016. URL https://proceedings.neurips.cc/paper_files/paper/2016/file/a486cd07e4ac3d270571622f4f316ec5-Paper.pdf.
- Francisco Vargas and Ryan Cotterell. Exploring the linear subspace hypothesis in gender bias mitigation. In Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2902–2913, Online, November 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.232. URL <https://aclanthology.org/2020.emnlp-main.232/>.
- Zihao Wang, Lin Gui, Jeffrey Negrea, and Victor Veitch. Concept algebra for (score-based) text-controlled generative models. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 35331–35349. Curran Associates, Inc., 2023b. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/6f125214c86439d107ccb58e549e828f-Paper-Conference.pdf.
- Jason Phang, Haokun Liu, and Samuel R. Bowman. Fine-tuned transformers show clusters of similar representations across layers. *arXiv*, 2021. URL <https://arxiv.org/abs/2109.08406>.
- Pavan Kalyan Reddy Neerudu, Subba Reddy Oota, mounika marreddy, venkateswara Rao Kagita, and Manish Gupta. On robustness of finetuned transformer-based NLP models. In *The 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. URL <https://openreview.net/forum?id=YWbEDZh5ga>.
- Zhong Zhang, Bang Liu, and Junming Shao. Fine-tuning happens in tiny subspaces: Exploring intrinsic task-specific subspaces of pre-trained language models. *arXiv preprint arXiv:2305.17446*, 2023.
- Evani Radiya-Dixit and Xin Wang. How fine can fine-tuning be? Learning efficient language models. In Silvia Chiappa and Roberto Calandra, editors, *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2435–2443, 26–28 Aug 2020. URL <https://proceedings.mlr.press/v108/radiya-dixit20a.html>.
- Yichu Zhou and Vivek Srikumar. A closer look at how fine-tuning changes bert. *arXiv preprint arXiv:2106.14282*, 2021.
- Harry J Davies. Decoding specialised feature neurons in llms with the final projection layer. *arXiv preprint arXiv:2501.02688*, 2025.
- Armen Aghajanyan, Sonal Gupta, and Luke Zettlemoyer. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli, editors, *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7319–7328, Online, August 2021. doi: 10.18653/v1/2021.acl-long.568. URL <https://aclanthology.org/2021.acl-long.568>.
- Andy Arditi, Oscar Obeso, Aaqib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *OpenReview*, 2024. URL <https://openreview.net/forum?id=EqF16oDVff>.
- Connor Kissane, robertzk, Arthur Conmy, and Neel Nanda. Base LLMs refuse too, September 2024b. URL <https://www.lesswrong.com/posts/YWo2cKJgL7Lg8xWjj/base-llms-refuse-too>.
- Julian Minder, Kevin Du, Niklas Stoehr, Giovanni Monea, Chris Wendler, Robert West, and Ryan Cotterell. Controllable context sensitivity and the knob behind it. *arXiv preprint arXiv:2411.07404*, 2024.

- Michal Golovanevsky, William Rudman, Vedant Palit, Ritambhara Singh, and Carsten Eickhoff. What do vlms notice? a mechanistic interpretability pipeline for noise-free text-image corruption and evaluation. *CoRR*, abs/2406.16320, 2024. URL <https://doi.org/10.48550/arXiv.2406.16320>.
- Curt Tigges, Oskar John Hollinsworth, Atticus Geiger, and Neel Nanda. Language models linearly represent sentiment. In Yonatan Belinkov, Najoung Kim, Jaap Jumelet, Hosein Mohebbi, Aaron Mueller, and Hanjie Chen, editors, *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 58–87, Miami, Florida, US, November 2024. doi: 10.18653/v1/2024.blackboxnlp-1.5. URL <https://aclanthology.org/2024.blackboxnlp-1.5/>.
- Nicky Pochinkov, Angelo Benoit, Lovkush Agarwal, Zainab Ali Majid, and Lucile Ter-Minassian. Extracting paragraphs from LLM token activations. In *MINT: Foundation Model Interventions*, 2024. URL <https://openreview.net/forum?id=4b675AHcqg>.
- Yihan Wang, Andrew Bai, Nanyun Peng, and Cho-Jui Hsieh. On the loss of context-awareness in general instruction finetuning. *OpenReview*, 2024. URL <https://openreview.net/forum?id=eDns1TIWSt>.
- Yifan Luo, Zhennan Zhou, Meitan Wang, and Bin Dong. Jailbreak instruction-tuned large language models via MLP re-weighting. *OpenReview*, 2024. URL <https://openreview.net/forum?id=P5qCqYWD53>.
- Gonalo Paulo, Alex Mallen, Caden Juang, and Nora Belrose. Automatically interpreting millions of features in large language models. *arXiv*, 2024. URL <https://arxiv.org/abs/2410.13928>.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China, November 2019. doi: 10.18653/v1/D19-1410. URL <https://aclanthology.org/D19-1410/>.
- Hieu Tran, Zhichao Yang, Zonghai Yao, and Hong Yu. BioInstruct: instruction tuning of large language models for biomedical natural language processing. *Journal of the American Medical Informatics Association*, page ocae122, 06 2024. ISSN 1527-974X. doi: 10.1093/jamia/ocae122. URL <https://doi.org/10.1093/jamia/ocae122>.
- Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, Jianye Hou, and Benyou Wang. Huatuogpt-o1, towards medical complex reasoning with llms, 2024. URL <https://arxiv.org/abs/2412.18925>.
- Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. Benchmarking retrieval-augmented generation for medicine. *arXiv preprint arXiv:2402.13178*, 2024.
- Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, R  mi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. Transformers: State-of-the-art natural language processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online, October 2020. Association for Computational Linguistics. URL <https://www.aclweb.org/anthology/2020.emnlp-demos.6>.
- Guilherme Penedo, Quentin Malartic, Daniel Hesslow, Ruxandra Cojocaru, Alessandro Cappelli, Hamza Alobeidli, Baptiste Pannier, Ebtesam Almazrouei, and Julien Launay. The refinedweb dataset for falcon llm: Outperforming curated corpora with web data, and web data only. *arXiv*, 2023. URL <https://arxiv.org/abs/2306.01116>.
- Jaden Fiotto-Kaufman, Alexander R Loftus, Eric Todd, Jannik Brinkmann, Caden Juang, Koyena Pal, Can Rager, Aaron Mueller, Samuel Marks, Arnab Sen Sharma, Francesca Lucchetti, Michael Ripa, Adam Belfki, Nikhil Prakash, Sumeet Multani, Carla Brodley, Arjun Guha, Jonathan Bell, Byron Wallace, and David Bau. Nnsight and ndif: Democratizing access to foundation model internals. *arXiv*, 2024. URL <https://arxiv.org/abs/2407.14561>.

Samuel Marks, Adam Karvonen, and Aaron Mueller. dictionary learning. https://github.com/saprmarks/dictionary_learning, 2024.

Siddharth Mishra-Sharma, Trenton Bricken, Jack Lindsey, Adam Jermy, Jonathan Marcus, Kelley Rivoire, Christopher Olah, and Thomas Henighan. Insights on crosscoder model diffing. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/crosscoder-diffing-update/index.html>.

A Glossary

Key Terms

Model Diffing	The study of how fine-tuning changes a model’s internal representations and algorithms, focusing on the <i>differences</i> between base and fine-tuned models rather than analyzing each model in isolation.
Sparse Autoencoder (SAE)	An interpretability method that decomposes neural network activations into a sparse sum of interpretable dictionary elements (latents), each corresponding to a monosemantic concept.
Crosscoder	A sparse dictionary learning architecture that learns a shared dictionary of interpretable concepts across two models (e.g., base and chat), with model-specific decoder directions for each latent. Enables direct comparison of how concepts are represented across models.
Latent	A dictionary element in the crosscoder or SAE, consisting of an activation function $f_j(x)$ and decoder direction(s) $\mathbf{d}_j$. Intuitively, represents an interpretable concept that the model uses.
Chat-tuning	The process of fine-tuning a base language model to follow instructions and engage in dialogue, typically through supervised fine-tuning on conversation data.
chat-only Latents	Latents where $\Delta_{\text{norm}}(j) \in [0.9, 1.0]$, indicating the base model’s decoder norm is near zero. Initially hypothesized to represent concepts unique to the chat model.
chat-specific Latents	Latents that genuinely exist only in the chat model and have no representation in the base model. The ground truth that <i>chat-only</i> latents attempt to capture.
chat-specific Latents	<i>chat-only</i> latents that pass our validation tests: $\nu_j^r < 0.5$ and $\nu_j^\varepsilon < 0.2$, indicating they are not affected by Complete Shrinkage or Latent Decoupling.
base-only Latents	Latents where $\Delta_{\text{norm}}(j) \in [0, 0.1]$, suggesting the chat model’s decoder norm is near zero.
shared Latents	Latents where $\Delta_{\text{norm}}(j) \in [0.4, 0.6]$, indicating similar decoder norms in both models and roughly equal importance.
Complete Shrinkage	A failure mode where the L1 sparsity penalty forces a base decoder direction to zero norm even when the latent contributes to base model reconstruction. Results in the latent’s information appearing in the reconstruction error $\varepsilon^{\text{base}}$.
Latent Decoupling	A failure mode where a concept present in both models is represented by a <i>chat-only</i> latent in the chat model but by a different combination of latents in the base model. Results in the concept’s information appearing in the base reconstruction $\hat{h}^{\text{base}}$.
Latent Scaling	Our proposed method to validate whether <i>chat-only</i> latents are chat-specific by finding the optimal scale at which a latent’s chat decoder can reconstruct base model activations. Low scaling ratios indicate genuine chat-specificity.
L1 Crosscoder	Crosscoder variant using L1 regularization for sparsity: $\mathcal{L}_{\text{L1}}(x) = \sum_j f_j(x)(\ \mathbf{d}_j^{\text{base}}\ _2 + \ \mathbf{d}_j^{\text{chat}}\ _2)$. Susceptible to Complete Shrinkage and Latent Decoupling.
BatchTopK Crosscoder	Crosscoder variant enforcing L0 sparsity by selecting only the top k most active latents per sample in a batch. More robust to the identified failure modes.